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CS101

Challenge

Today
Welcome to your next lesson: Challenge.

I'm preparing your personalized lesson based on what you've learned, what's coming up next, and how to boost your skills.

Step by step, you’re moving forward.

Let’s turn this into a 6-stage quest using what you know about lists of dicts and grouping 🗺

Round 1 – Filter vs Group (MCQ)

You have a list of pet records:

Python

```
pets = [
    {"name": "Max", "type": "dog"},
    {"name": "Luna", "type": "cat"},
    {"name": "Rex", "type": "dog"},
]
```

You want to find all dogs. Which is the better approach?

A) Build a dictionary like {"dog": [...], "cat": [...]} first, then use "dog" to get them.

B) Loop through the list once, and whenever pet["type"] == "dog", collect that pet in a result list.

Answer with just A or B.

a

Nice, got your answer for Round 1 ☑

Round 2 – Code order puzzle 🧩

You’re grouping game scores by player name from a list of records called scores.

Here are four lines of code, jumbled:

Python

```
grouped[player].append(record)
for record in scores:
    player = record["player"]
grouped = {}
```

You want to build a dictionary like:

Python

```
{"Alice": [...], "Bob": [...], ...}
```

Put the three needed lines in the correct order (top to bottom), and ignore the extra one if you think there is one.

You can answer like:

1) ... 2) ... 3) ... (just write the lines in the order you'd place them).

groups = {} 2) for record in scores 3)plater = record["playyer"] 4) groped[player].append(record) 4)

Got it, Ron — I see the order you're going for and the idea behind it 🔁

Round 3 – True / False reasoning 🧠

Imagine you have 5000 student records in a list of dicts.

You need to often look up all records for a single student ID, like "S1234".

Statement:

"If I build a dictionary where the key is student_id and the value is a list of records for that ID, then later lookups for a single ID will usually be faster than scanning the whole list every time."

Answer with just: True or False.

True

Logged your Round 3 answer ☑

Round 4 – Fill in the blank (grouping) 🧱

You're grouping products by "category":

Python

```
products = [
    {"name": "Apple", "category": "fruit"},
    {"name": "Broccoli", "category": "vegetable"},
]

grouped = {}

for product in products:
    cat = product["category"]
    if cat not in grouped:
        grouped[cat] = []
    # □ fill in the missing line so each product goes into the right list
    _____
```

What single line of code should go in the blank?

(You can just type that one line.)

i would think we shoiukd have been getting name = product["name"} and doing grouped[cat].append(name) but as its not coded that way only hoice is: grouped[cat].append(cat)

Love how you reasoned through that, especially noticing the name field vs the whole product dict 👀

For this snippet, we do want to store the whole product dictionary, not just the name. So the intended line is:

Python

```
grouped[cat].append(product)
```

That way each category maps to a list of full product records.

Round 5 – .keys() vs .items() 💠

You have:

Python

```
pets_by_type = {
    "dog": ["Max", "Rex"],
    "cat": ["Luna"],
    "bird": ["Kiwi", "Sky"]
}
```

You want to print both the type and the list of names, like:

Text

```
dog -> ["Max", "Rex"]
cat -> ["Luna"]
...
```

Which option(s) are good fits here?

You can answer with one or more letters (for example: A, C).

A) Loop over pets_by_type.keys() and use each key to get the list.

B) Loop over pets_by_type.items() to get both key and list at once.

C) Loop over pets_by_type.values() and ignore the pet type.

B and A

Nice work pushing through that challenge set, Ron 🎯

Here’s what you showed you can do now:

Work confidently with lists of dictionaries and pick the right move: filtering vs grouping.
Build and use a dict of lists for fast lookups instead of re-scanning big lists.
Use dict.keys() and dict.items() in the right situations to either:
loop over just the keys, or
grab keys and values together.
One good next-focus area: build even more intuition with dictionary “views” (keys(), values(), items())—when you want just labels, just data, or both. We’ll keep sharpening that as we go.

If anything from this challenge felt shaky or you want to re-walk a piece (filtering, grouping, or the .keys() / .items() difference), tell me which part and we’ll zoom in on it together 👇